Akoizo Original Practice - Form 4

Division C | Engineering Cad | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (4 points) Calculate the base volume before cutting the hole.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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2. (4 points) Calculate the material removed by the through hole.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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3. (4 points) Calculate the remaining base volume.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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4. (4 points) Calculate the hole diameter.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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5. (4 points) Calculate the solid base centroid x coordinate after the centered hole is cut.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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6. (4 points) Calculate the solid base centroid y coordinate.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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7. (4 points) Calculate the solid base centroid z coordinate.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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8. (4 points) Calculate the separate two-component assembly center-of-mass x coordinate.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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9. (4 points) Calculate the separate assembly center-of-mass y coordinate.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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10. (4 points) A second version doubles every linear dimension of the perforated base. By what factor does its volume increase?

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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11. (4 points) Convert the base length to centimetres.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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12. (4 points) Describe a feature sequence that makes the base and centered hole, including a reliable end condition for the cut.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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13. (4 points) Why can a sketch with the correct visible shape still be under-constrained? Name a way to remove an unintended degree of freedom.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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14. (4 points) Why is the arithmetic midpoint of the assembly positions generally the wrong center of mass for this case?

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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15. (4 points) Explain how two parts can have equal volume but different center-of-mass positions.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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16. (4 points) A reported part volume is numerically correct in cm^3 but entered as mm^3. Explain the scale error.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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17. (4 points) Describe how to check that a hole is a cut rather than a separate overlapping solid.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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18. (4 points) Explain why matching the part volume alone does not prove that an assembly is correct.

Authored CAD case 4. Units are mm. A rectangular base is L=90, W=40, H=14, occupying x=0..90, y=0..40, z=0..14. A cylindrical hole of radius 6 goes through the full height, centered at (x=45, y=20). Ignore chamfers and fillets. Use pi=3.141592653589793. A separate assembly has point-mass components of mass 180 g at (40,0,0) mm and 100 g at (60,0,0) mm. Do not infer any other component or coordinate.

Answer: _____
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